Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	5 july 2026
Team ID	
Project Name	Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

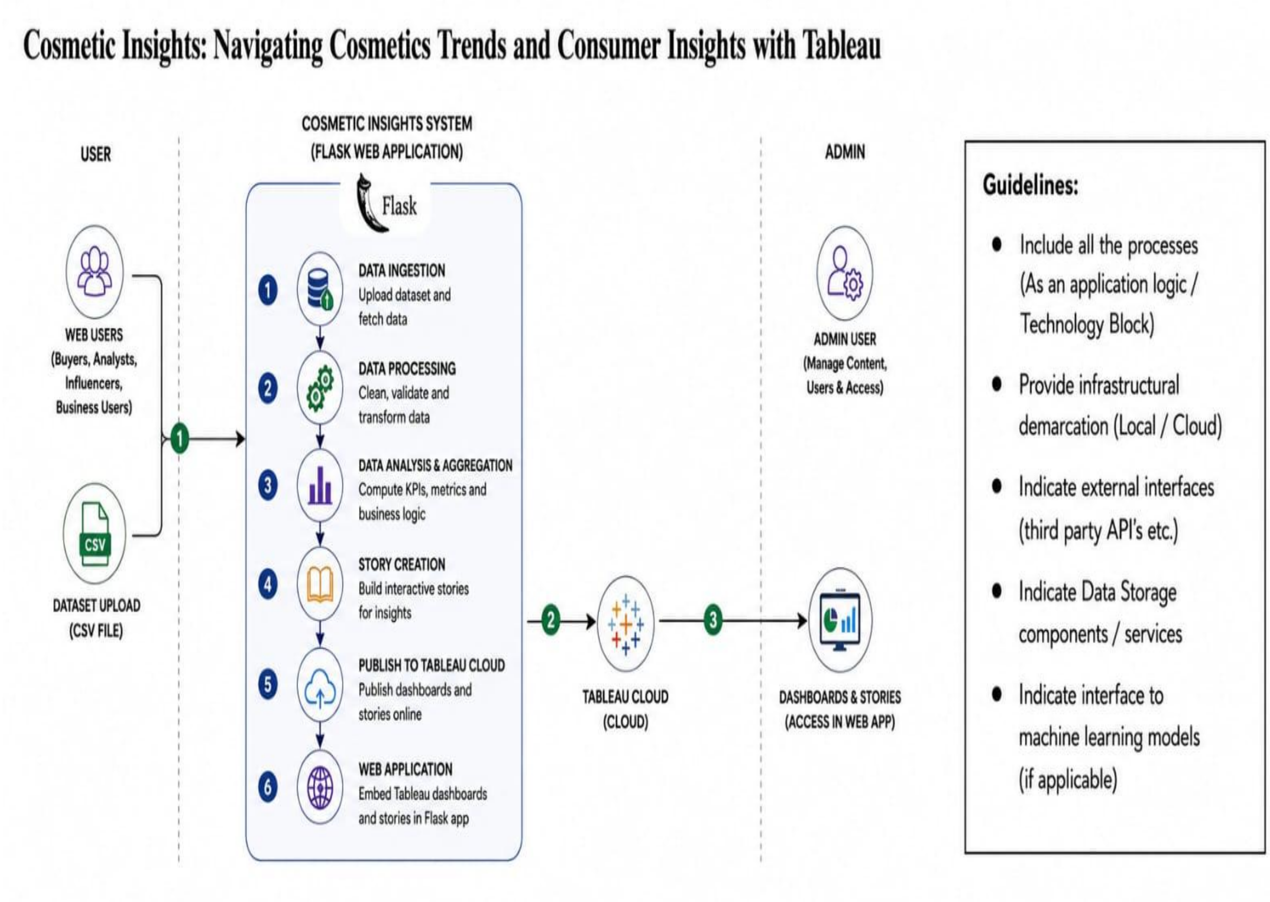

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	Data Source	Heart disease dataset containing patient demographics, lifestyle, and clinical information.	CSV Dataset
2	Data Preparation	Cleaning, formatting, and preprocessing of the dataset before visualization.	Tableau Desktop
3	Calculated Fields	Creation of custom measures and KPIs for enhanced analysis.	Tableau Calculated Fields
4	Data Visualization	Development of charts, graphs, and visual analytics.	Tableau Desktop
5	Dashboard Development	Creation of an interactive dashboard with filters and KPI cards.	Tableau Dashboard
6	Story Development	Presentation of analytical insights using sequential story points.	Tableau Story
7	Dashboard Publishing	Publishing dashboards and stories for online accessibility.	Tableau Public
8	User Interface	Responsive web interface to access dashboards, stories, project information, and developer details.	HTML5, CSS3, Bootstrap, JavaScript
9	Application Logic	Handles webpage routing and embeds Tableau Dashboard and Story within the application.	Python, Flask
10	File Storage	Stores datasets, images, and project assets locally.	Local File System
11	Infrastructure (Server)	Local deployment and execution of the web application.	Flask Development Server

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	Open-source frameworks and tools used to develop the application.	Flask, Bootstrap, Tableau Public
2	**Security Implementations**	Client-side validation, secure routing, and safe embedding of Tableau visualizations.	HTML5 Validation, Flask Routing
3	**Scalable Architecture**	Modular architecture separating data, visualization, and presentation layers.	Three-Tier Architecture (Data Layer, Visualization Layer, Presentation Layer)
4	**Availability**	Dashboard and Story published online and accessible through the web application.	Tableau Public, Flask
5	**Performance**	Optimized dashboards, responsive frontend, and lightweight web application for faster loading.	Tableau Public, Bootstrap, JavaScript